Mechanical waves and distant effects

antocciphysics | TEACHER KEY | CPA

Learning target: Use travel-time reasoning while distinguishing waves from the motion of material.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 6, Day 14; 3 course days later.*
A diverging lens spreads initially parallel rays outward.

02. Explain a worked example (worked)
Half amplitude; quarter intensity under the stated same-medium model.

03. Look and explain (model)
Disturbance propagates while fluid motion is mostly local oscillation, with possible transport not identical to wave speed.

04. Build the solution (practice)
10 s.

05. Use the idea (reason)
Need amplitude/energy, frequency, site conditions, geometry, structures, bathymetry for water waves, etc.

06. Choose a method (mixed) *Review: Unit 6, Day 5; 12 course days later.*
Away from the normal; frequency stays fixed at a stationary boundary.

07. Exit check (exit)
Doubles.

08. STRETCH: Apply it another way (stretch)
Ratio 9: 1; absorption, reflection, terrain, source direction or local coupling.

Mechanical waves and distant effects

antocciphysics | TEACHER KEY | HONORS

Learning target: Use travel-time reasoning while distinguishing waves from the motion of material.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 6, Day 14; 3 course days later.*
$d_i \approx -10$ cm; $m \approx +1.667$. Upright, enlarged virtual image.

02. Explain a worked example (worked)
Half amplitude; quarter intensity under the stated same-medium model.

03. Represent and interpret (model)
Disturbance propagates while fluid motion is mostly local oscillation, with possible transport not identical to wave speed.

04. Solve and justify (practice)
P 35 s; S 60 s; delay 25 s.

05. Analyze the reasoning (reason)
Need amplitude/energy, frequency, site conditions, geometry, structures, bathymetry for water waves, etc.

06. Choose a method (mixed) *Review: Unit 6, Day 5; 12 course days later.*
$\theta_c \approx 36.03°$ from the normal. Above it, total internal reflection occurs.

07. Exit check (exit)
31.6 m/s; 316 s; requires long shallow-water wavelength regime and simplified constant depth; not damage forecast.

08. STRETCH: Apply it another way (stretch)
8/9 W/m²; $P = I\,4\pi r^2 = 3200\pi$ W ≈ 10053 W. Ground-wave geometry, material and attenuation need different models.